
Chen-Li 4.7

2026-08-10

Danus arXiv DOI

J. Chen and J. Li, *Mean Curvature Flow of Surface in 4-Manifolds*, Adv. Math. 163 (2001), 287-309, Theorem 4.7

Chen-Li 4.7

1. Wang (2001) Kähler-Einstein Li-Yang (2014) Zhang
(2015) Qiu-Sun (2019/2020) Kähler

2. Chen-Li (2002) Remark 6.7 “
”

3. Han-Li Han-Li-Sun Li-Sun

4. Chen-Li (2004) Neves (2007) Maslov

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Wang “ ” Chen-Li (2002)

1.

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- Chen-Li 4.7
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- **A**
- **B**

[arXiv](#)

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$$\sup_{\Sigma_t} |A|^2 \leq \frac{C}{T-t}, \qquad t < T.$$

2.

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2000	Chen-Tian, <i>Moving Symplectic Curves in Kähler-Einstein Surfaces</i> [R1]	Theorem 2.4 Kähler Theorem 2.5	A
2001	Wang, <i>Mean Curvature Flow of Surfaces in Einstein Four-Manifolds</i> [R3]	Theorem A Kähler-Einstein Kähler Huisken	Chen-Li A
2002	Chen-Li, <i>Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces</i> [R4]	Theorems 1.2, 4.3, 5.1, 6.6 Remark 6.7	“ ” A
2014	Li-Yang, <i>Symplectic Mean Curvature Flows in Kähler Surfaces with Positive Holomorphic Sectional Curvatures</i> [R12]	2 Kähler Theorem 1.3 Chen-Li/Wang	Kähler-Einstein A
2014	Li-Yang, <i>Generalized Symplectic Mean Curvature Flows in Almost Einstein Surfaces</i> [R13]		MCF B
2015	Zhang, <i>Remarks on Symplectic Mean Curvature Flows...</i> [R14]	Li-Yang Kähler Theorem 3.2 /	arXiv [R12] A
2019	Qiu-Sun, <i>Mean Curvature Flow of Surfaces in a Hyperkähler 4-Manifold</i> [R15]	Theorem 1.2 Theorem 1.3	White A
2026	Chen-Han-Li-Sun, <i>Tangent Flows of Symplectic Mean Curvature Flows</i> [R18]	Type I*	“ ” “ ” B

3.

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2009	Han-Li, <i>Singularities of Symplectic and Lagrangian Mean Curvature Flows</i> [R5]	Theorem 2.4 $\mathbb{C}^2$ Corollary 2.6 $-2 \leq K \leq 0$	A
2009	Han-Li, <i>Translating Solitons to Symplectic and Lagrangian Mean Curvature Flows</i> [R6]	/	A
2011	Han-Li-Sun, <i>The Second Type Singularities...</i> [R8]	Theorems 1.1-1.4 Kähler	A
2024	Li-Sun, <i>Eternal Solutions to Almost Calibrated Lagrangian and Symplectic Mean Curvature Flows</i> [R16]	Kähler /	B

4.

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2010	Han-Li, <i>The Mean Curvature Flow Approach to the Symplectic Isotopy Problem</i> [R7]	Kähler-Einstein	A
2011	Han-Li, <i>Long Time Existence of the Symplectic Mean Curvature Flow</i> [R9]	Kähler	arXiv A
2013	Han-Li-Yang, <i>Symplectic Mean Curvature Flow in $\mathbb{CP}^2$</i> [R10]	$\mathbb{CP}^2$	A/B
2014	Han-Li, <i>The Mean Curvature Flow Along the Kähler-Ricci Flow</i> [R11]	Kähler-Ricci MCF /	A
2024	Han-Li-Sun, <i>Gradient Flow for β-Symplectic Critical Surfaces</i> [R17]	β - / β -	gradient-flow MCF A

5.

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2004	Chen-Li, <i>Singularity of Mean Curvature Flow of Lagrangian Submanifolds</i> [L1]	Calabi-Yau Theorem 1.1 2Corollary 6.7	A
2007	Neves, <i>Singularities of Lagrangian Mean Curvature Flow: Zero-Maslov Class Case</i> [L2]	Theorems A-BMaslov	Maslov “ ” A
2017	Sun-Yang, <i>Generalized Lagrangian Mean Curvature Flows in Almost Calabi-Yau Manifolds</i> [L3]	Neves LMCFCorollary 1.1	A
2021	Lambert-Lotay-Schulze, <i>Ancient Solutions in Lagrangian Mean Curvature Flow</i> [L4]	blow-down Lawlor neck blow-down	A
2024	Lotay-Schulze-Székelyhidi, <i>Ancient Solutions and Translators of Lagrangian Mean Curvature Flow</i> [L5]	blow-down MaslovBrakke	/ A
2025	Li-Luo-Sun, <i>Type II Singularities of Lagrangian Mean Curvature Flow with Zero Maslov Class</i> [L6]	Maslov	A

6.

Kähler	$\cos \alpha > 0$		[R1], [R2], [R3]
Huiskens	$1/\cos \alpha$	/	[R2], [R4], [R8]
	$ A ^2 \leq C/(T-t)$		[R4], [L1]
			[R15]
	$(T-t) \max A ^2 \rightarrow \infty$	/	[R5], [R6], [R8], [R16]
	$ A ^2, H ^2, \cos \alpha$		[R7], [R9], [R10], [R12], [R14]

7. 4.7

7.1

1. Wang Theorem A Chen-Li Theorem 4.7 Kähler-Einstein
2. Chen-Li 2002, Remark 6.7
3. Qiu-Sun Theorem 1.3 Kähler

7.2

- “ ” 1
- “ ” “ ” White
- Chen-Li 2002 “ ” Wang
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7.3

Chen-Li/Wang “ ”
White /
Chen-Li 2002 “ ”

8.

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- **Li-Sun 2024 [R16]**
- **Han-Li-Sun 2024 [R17]** β -
- **Li-Luo-Sun 2025 [L6]** Maslov
- **Chen-Han-Li-Sun 2026 [R18]** MCF Type I*

arXiv:2309.16478 2025-10-13

9.

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10.

Chen-Li	4.7	Wang 2001	Chen-Li 2002	Li-Yang 2014	Zhang 2015	Qiu-Sun
2019/2020						
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		/	Chen-Li 2002			
Kähler	Qiu-Sun					